

How to Measure Legislative District Compactness If You Only Know it When You See it¹

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¹Based on joint work with Aaron Kaufman and Mayya Komisarchik

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- Is this any way to run a democracy? Our discipline ought to help.

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 - Required in many other jurisdictions

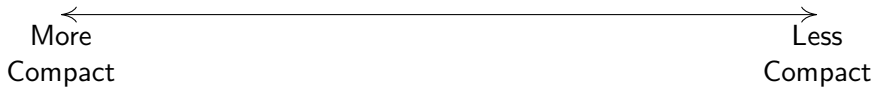
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A simple single compactness dimension that you know when you see

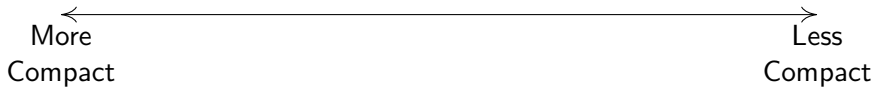
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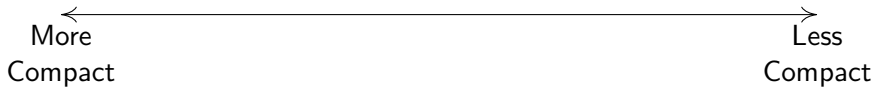
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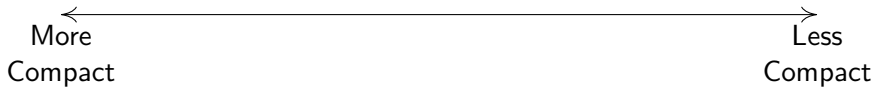
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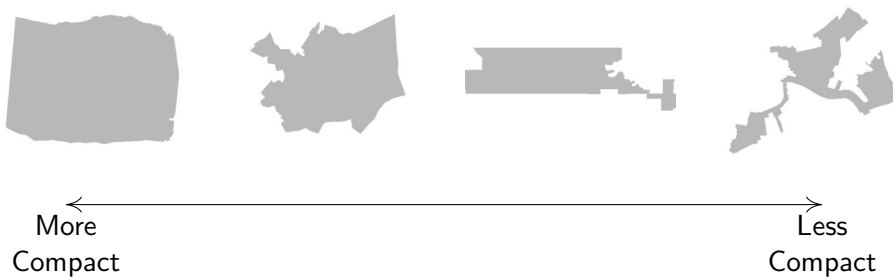
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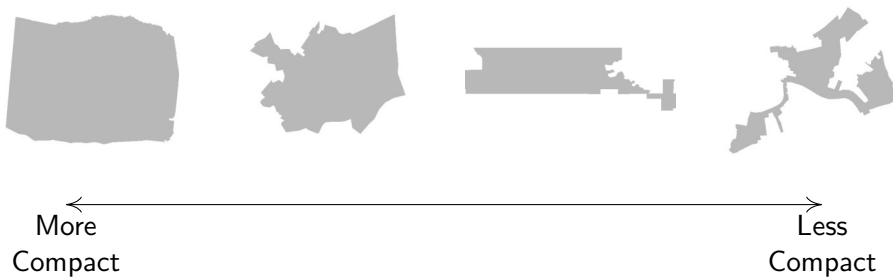
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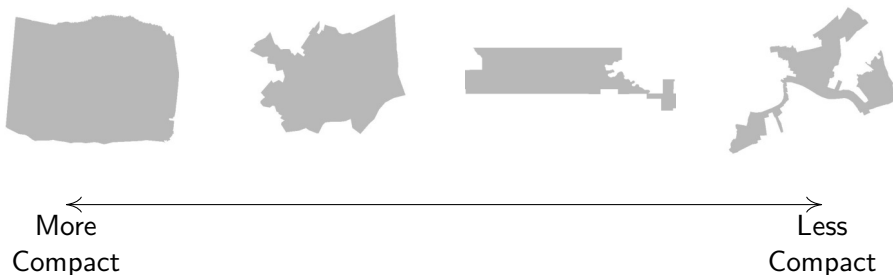
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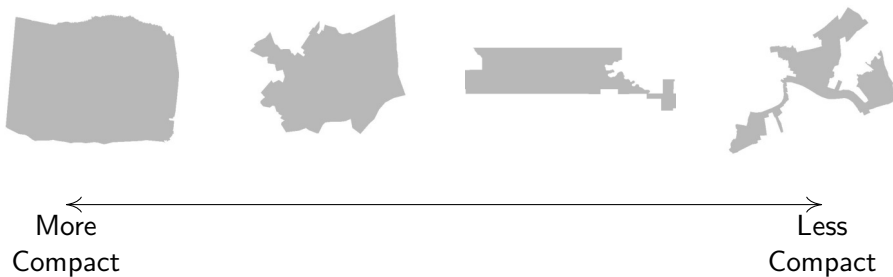
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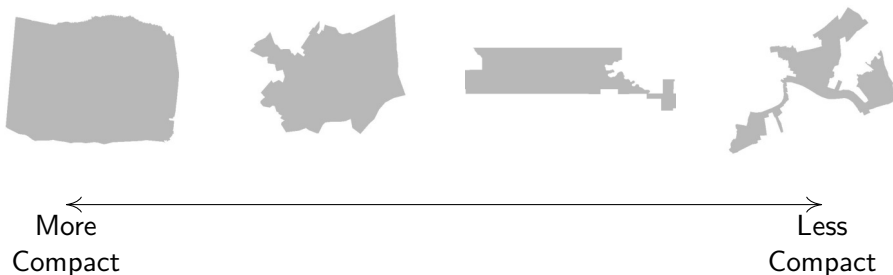
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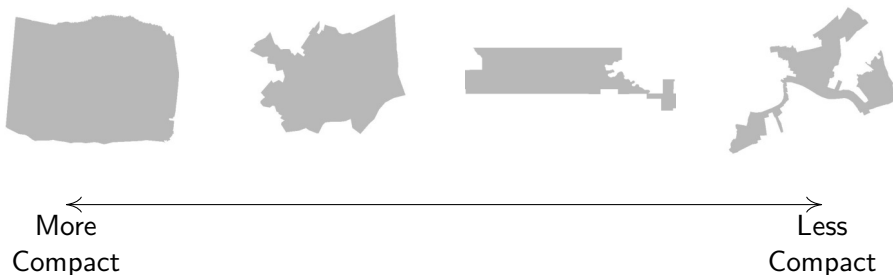
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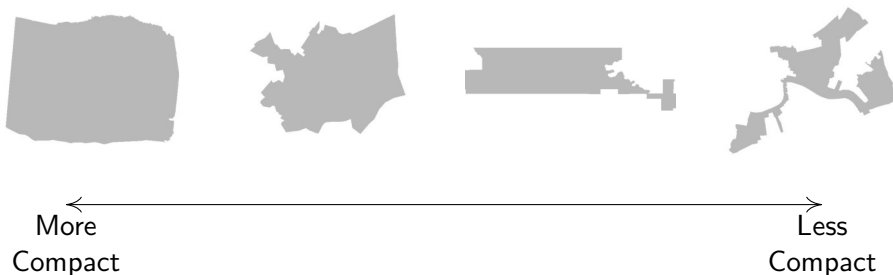
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- **↪ Let's start with existing measures by social scientists**

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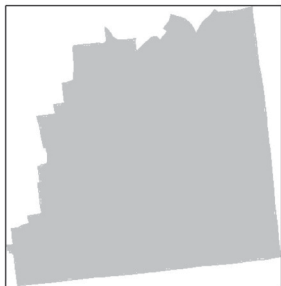
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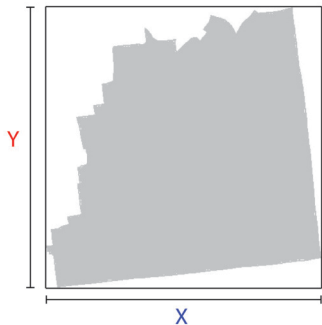
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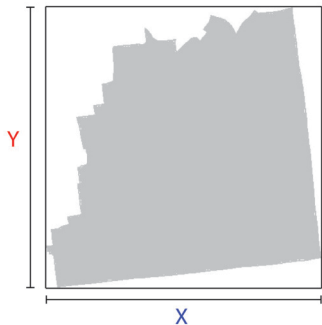
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In both districts: $X/Y \approx 1.30$

Measure 2: Reock, District / Bounding Circle Areas

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Circular districts are most compact

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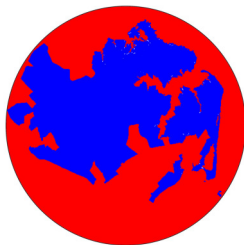
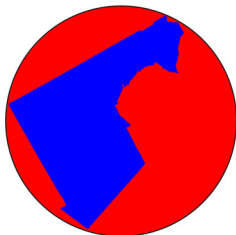
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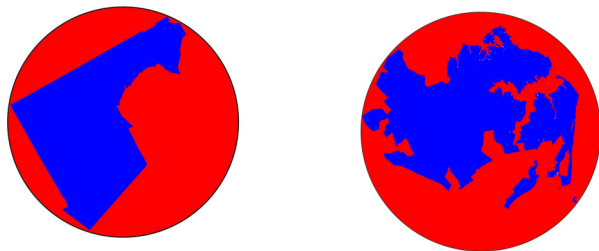
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In both cases, $X/(Y + X) \approx 0.37$

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In both cases, $\text{MAD}(r)/\bar{r} \approx 0.31$

A Brief Rotational Invariance Interlude:

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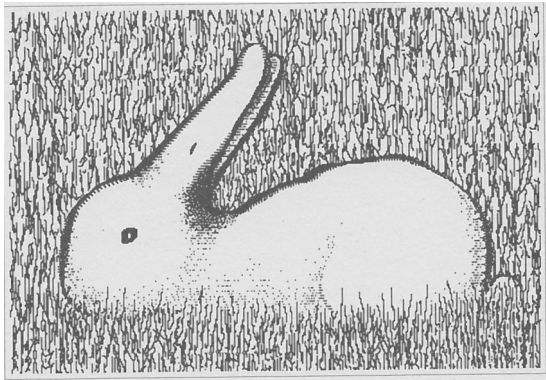
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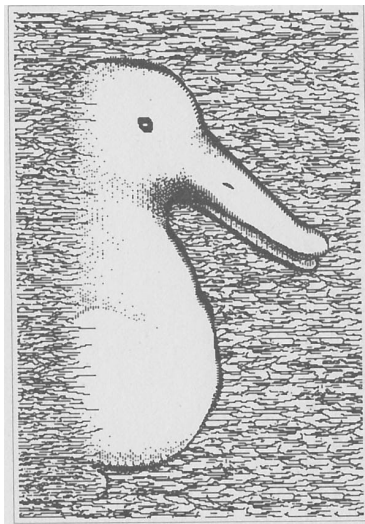
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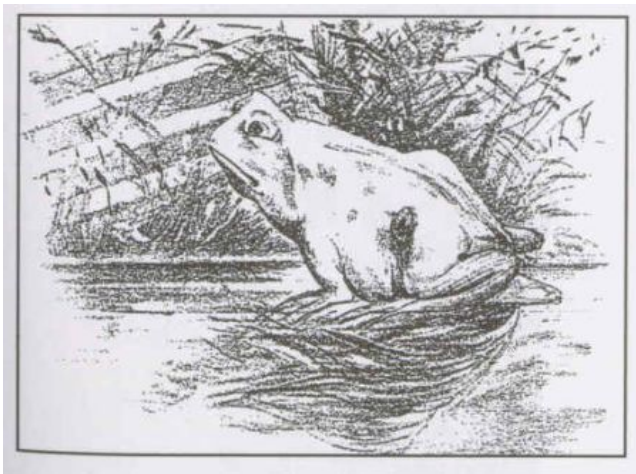
A Brief Interlude on Perception: See the Rabbit?



A Brief Interlude on Perception: See the Rabbit Duck?



A Brief Interlude on Perception: See the Frog?



A Brief Interlude on Perception: See the Frog Horse?



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- ↪ Measuring “you know it when you see it”: No rotational invariance

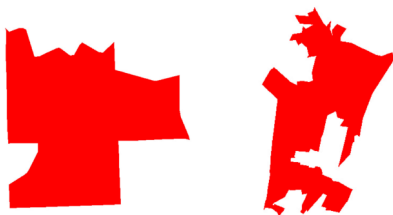
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Symmetric figures (circles, squares) are more compact

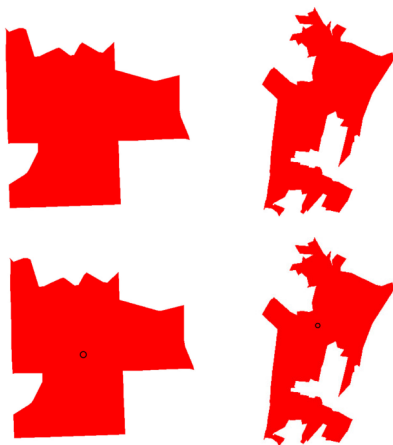
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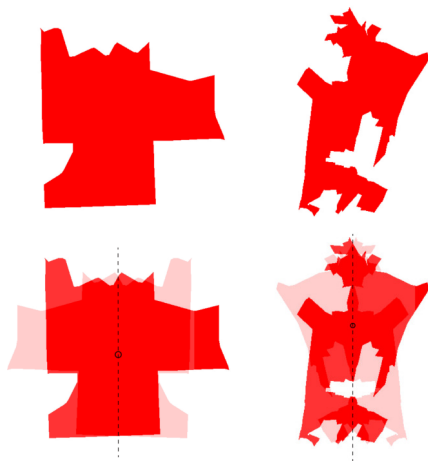
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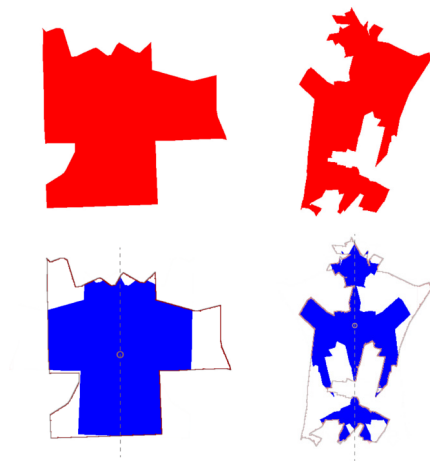
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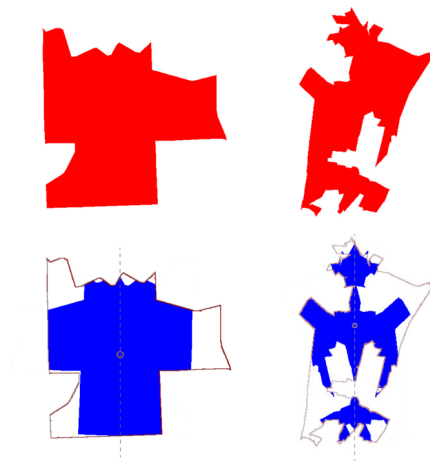
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In both cases, $\text{Overlap} / \text{Original Area} \approx 0.34$

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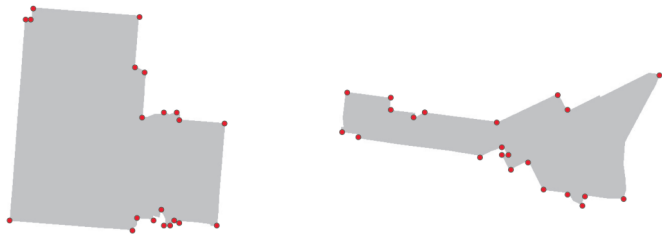
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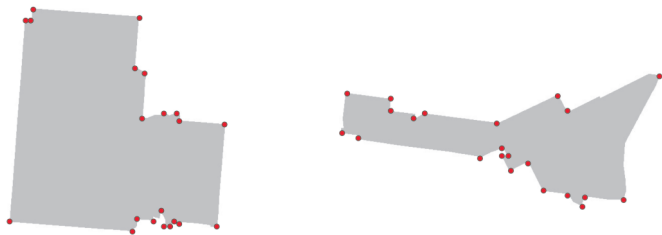
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Both districts have 21 significant corners

Which is more compact?



Which is more compact? Depends on the standard!



Which is more compact? Depends on the standard!



Convex Hull

4

3

2

1

Which is more compact? Depends on the standard!



	4	3	2	1
Convex Hull	4	3	2	1
Reock	1	2	3	4

Which is more compact? Depends on the standard!



Convex Hull	4	3	2	1
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- **Unusual?** From 18,215 Congressional and State Legislative Districts, we found 162 trillion others (about 0.15%)
- **Many more inconsistencies on individual districts**

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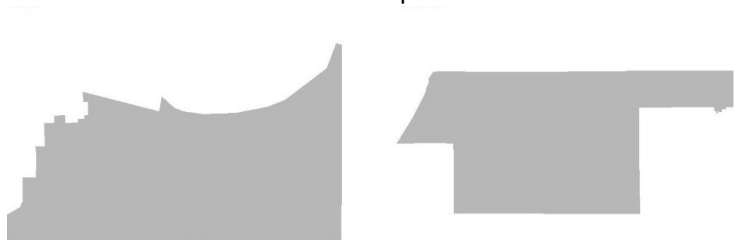
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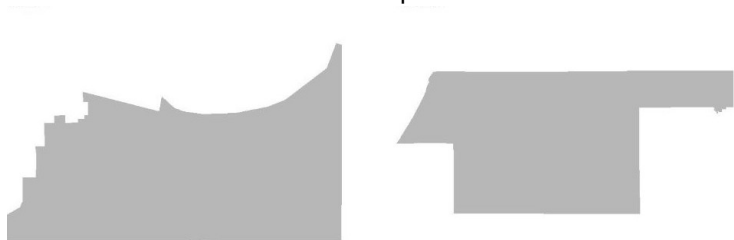
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Full Ranking

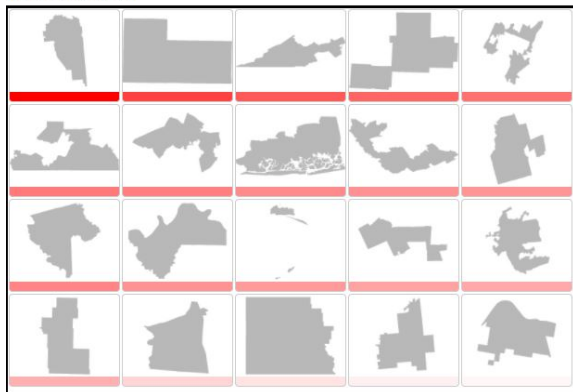


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MOST Compact Here



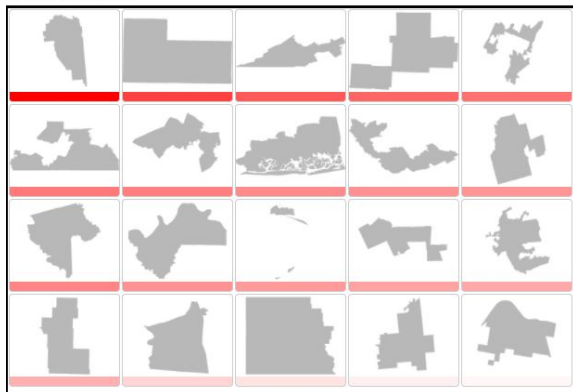
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We show: very high reliability

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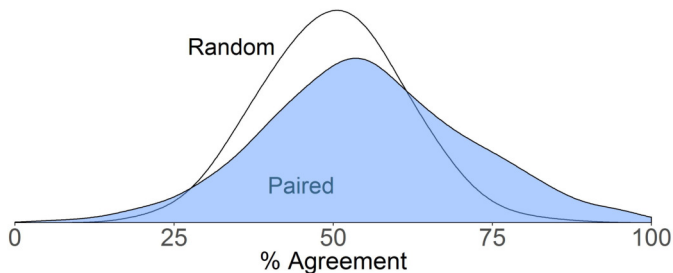
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Ranking: all evaluations on **one dimension** of user's choice

Intercoder Reliability of Pairs

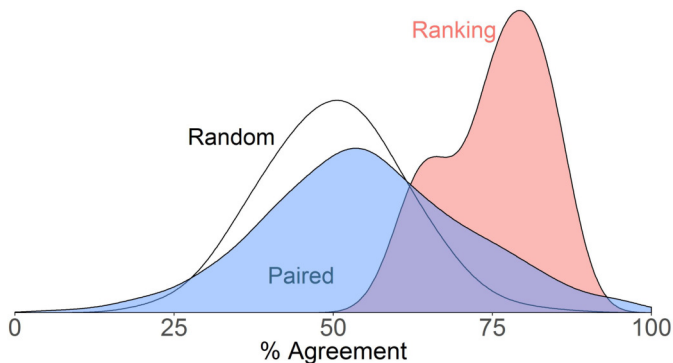
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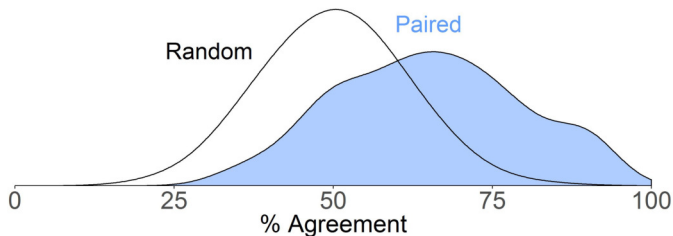
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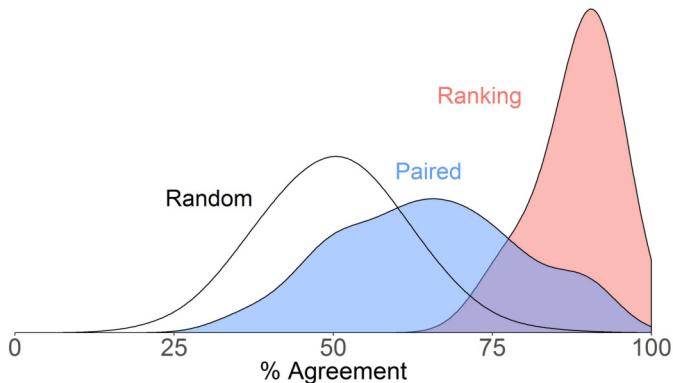
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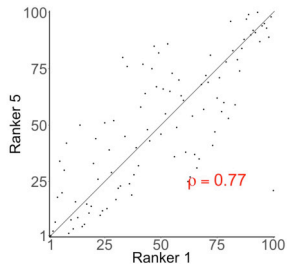


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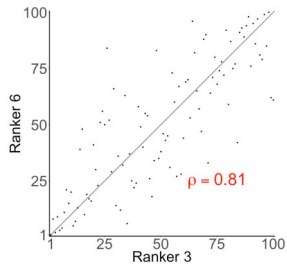
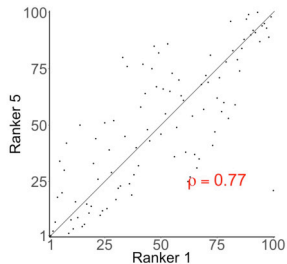
Paired Comparisons: better than chance; Pairs implied by ranks: much better



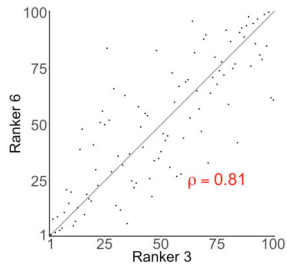
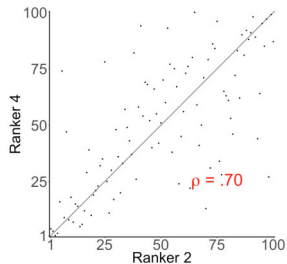
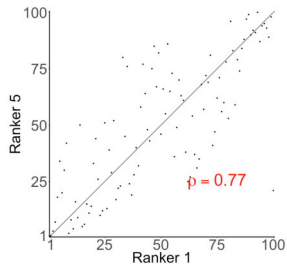
Intercoder Reliability on Ranks



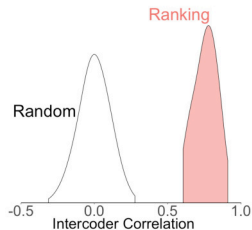
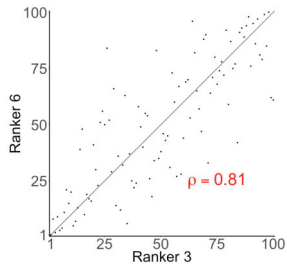
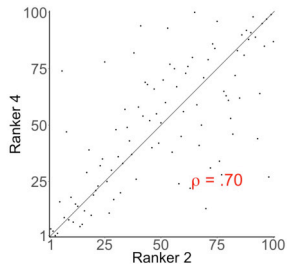
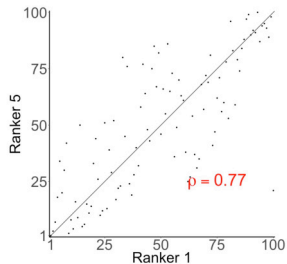
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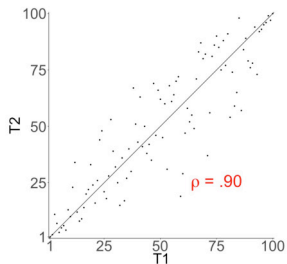
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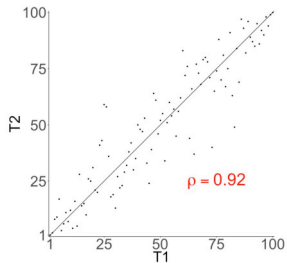
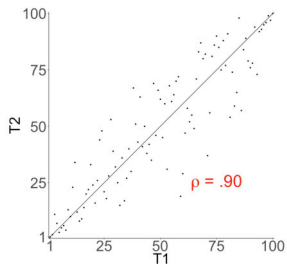
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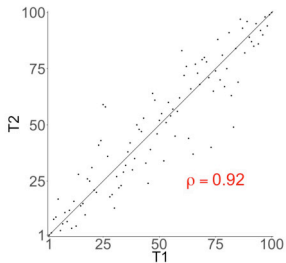
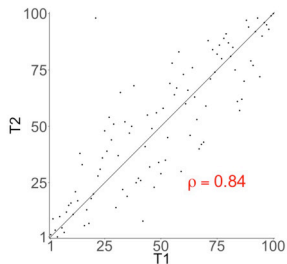
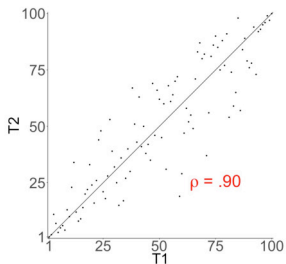
Intracoder Reliability on Ranks



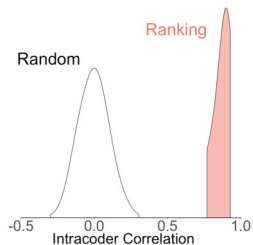
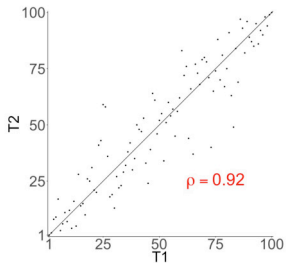
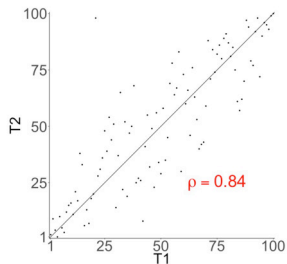
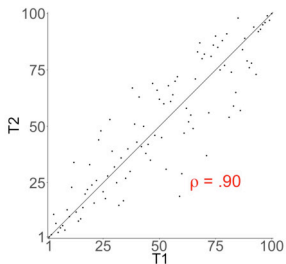
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 - (Not a description of any one existing measure)

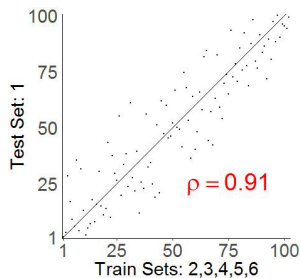
Model Validation: 6-Fold Cross-validation

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Predict Test Set from 5 Training Sets

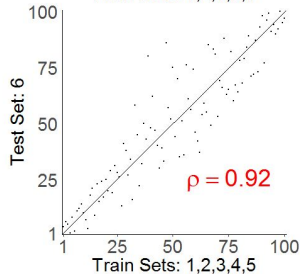
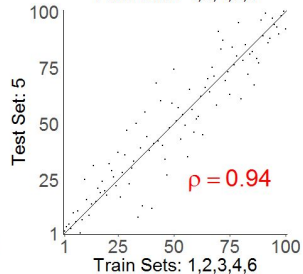
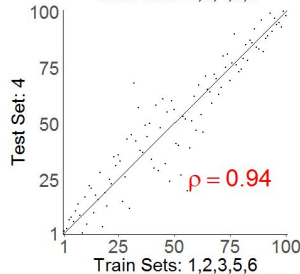
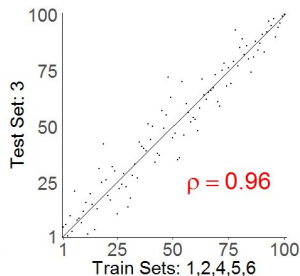
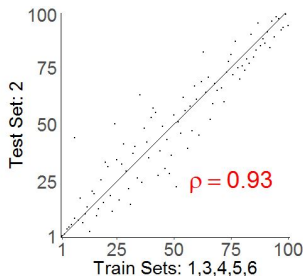
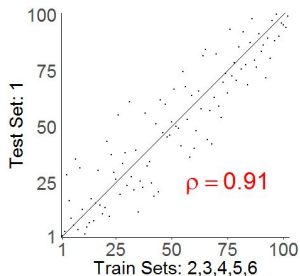
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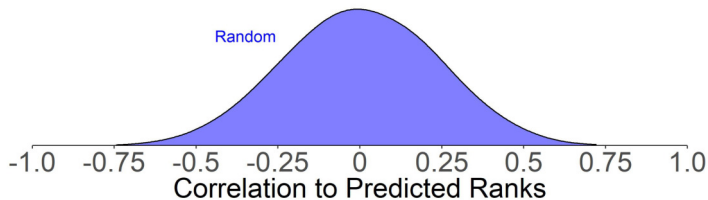
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Respondents ranging from ordinary citizens to those responsible for redistricting

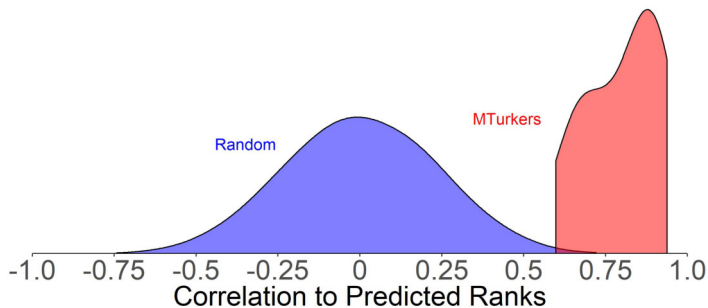
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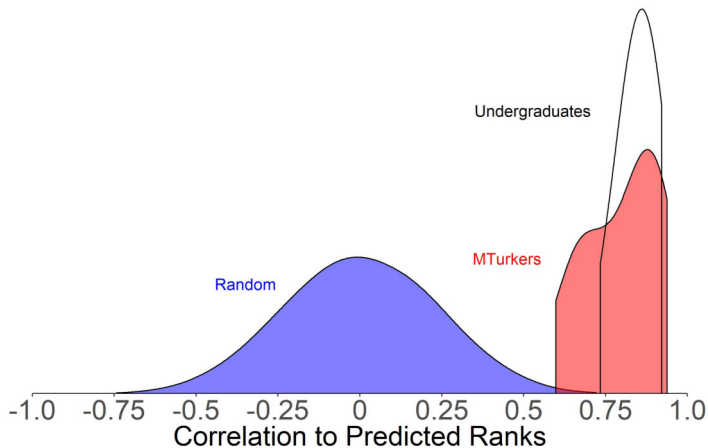
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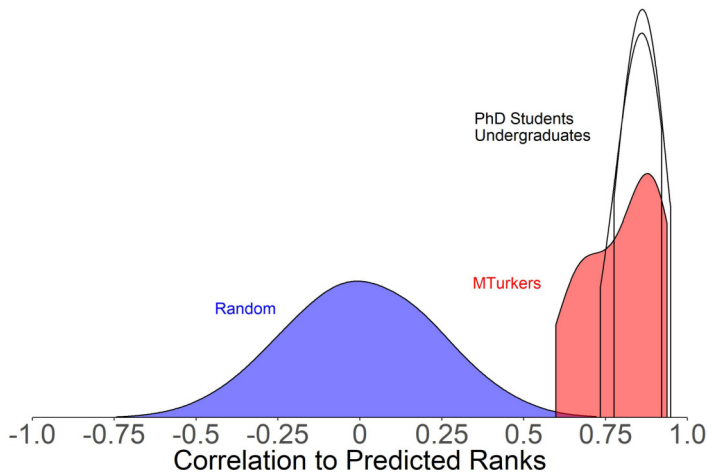
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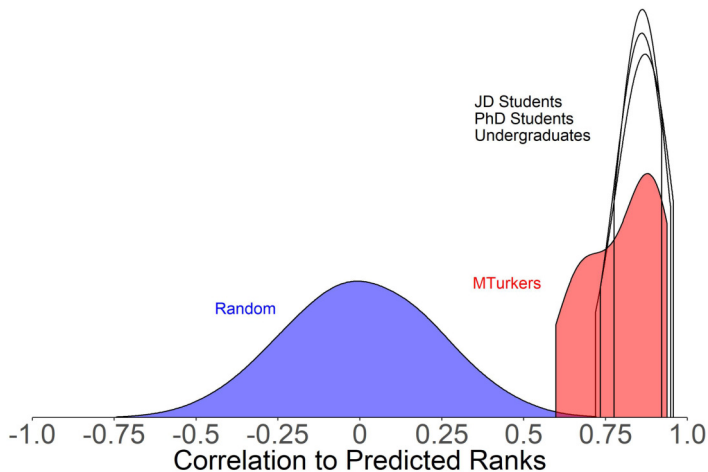
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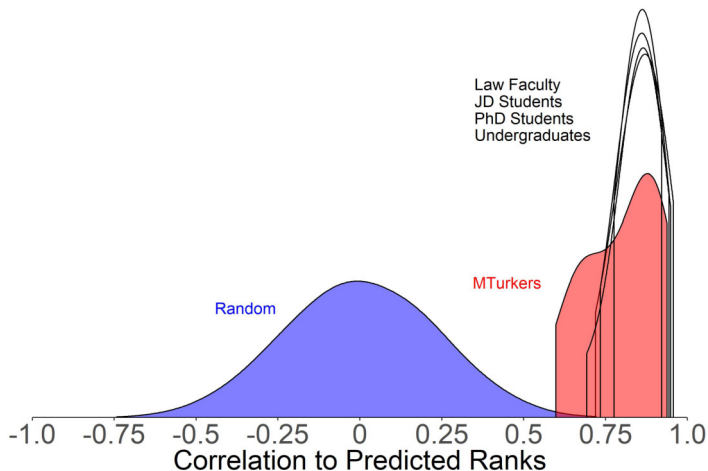
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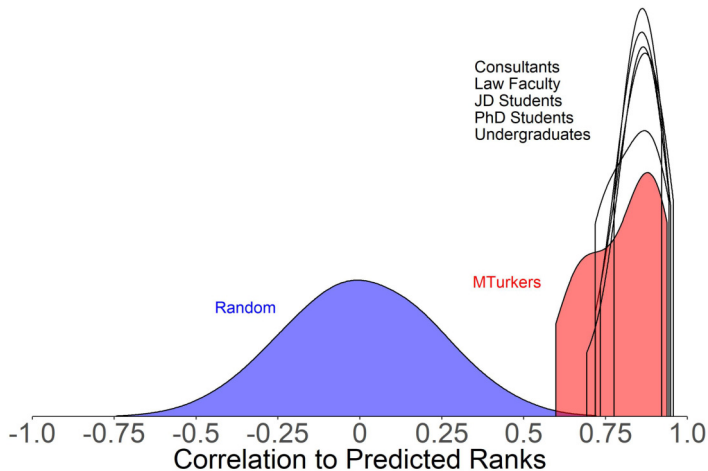
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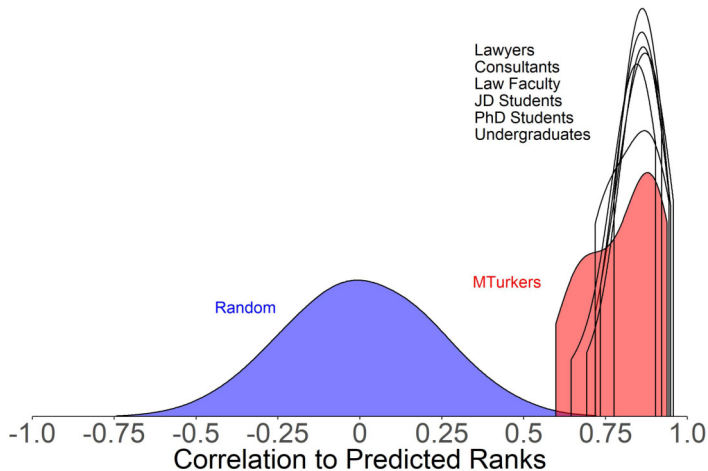
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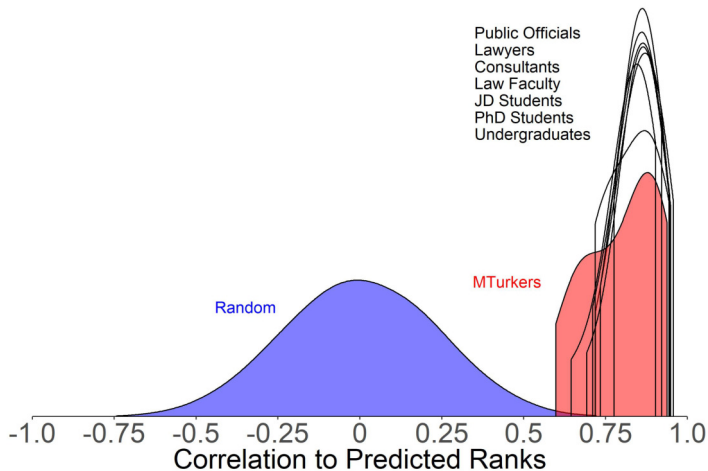
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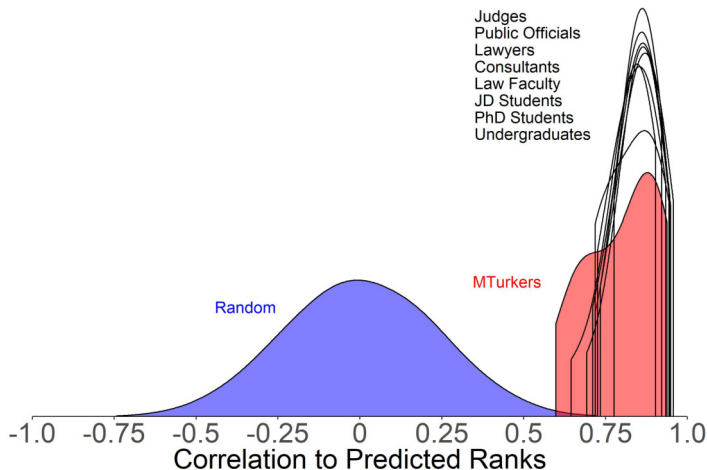
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COMPACT

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For more information



AaronRKaufman.com



GaryKing.org



MayyaKomisarchik.com

Paper, data, software, slides: j.mp/Compactness